

Submission CALC1-MID-SUB01

Question CALC1-MID-Q01

scratch: show step

1. $f'(x) = 2x(2x-1) + (x^2+4)^2$

2. $f'(1) = 2(1) + (5)^2 = 10$

=> FINAL: 10 (check units/interpretation)

Question CALC1-MID-Q02

scratch: show step

1. $\int 2x \, dx = (2/2)x^2$

2. $[2/2 x^2]_0^3 = 9$

=> FINAL: 9 (check units/interpretation)

Question CALC1-MID-Q03

scratch: show step

1. $2w+2l=22 \Rightarrow l=11-w$

2. $A(w) = w(11-w)$; at $w=8$, $A=24$

3. A maximum requires $A'(w)=0$, so the stated dimensions are not assumed optimal without that check.

=> FINAL: $A(w) = w(11-w)$; $A(8)=24$; check $A'(w)=0$ (check units/interpretation)

Question CALC1-MID-Q04

scratch: show step

1. $x^2-16=(x-4)(x+4)$

2. cancel $x-a$ only after identifying $x \neq a$

3. $\lim_{x \rightarrow 4} \frac{x^2-16}{x-4} = 8$

=> FINAL: 8 (check units/interpretation)

Question CALC1-MID-Q05

scratch: show step

1. Critical point: $f'(c)=0$ or $f'(c)$ is undefined (when f is defined).

2. A sign change in f' from + to – indicates a local maximum.

=> FINAL: critical point: $f'(c)=0$ or undefined; + to – sign change gives a local maximum (check units/interpretation)

Question CALC1-MID-Q06

scratch: show step

1. $f'(x) = 2x(2x-1) + (x^2+2)^2$

2. $f'(3) = 6(5) + (11)^2 = 14$

=> FINAL: 14 (check units/interpretation)

Question CALC1-MID-Q07

scratch: show step

1. $\int 2x \, dx = (2/2)x^2$

2. $\left[\frac{2}{2} x^2\right]_0^4 = 16$

=> FINAL: 16 (check units/interpretation)

Question CALC1-MID-Q08

scratch: show step

1. $2w+2l=24 \Rightarrow l=12-w$

2. $A(w)=w(12-w)$; at $w=9$, $A=27$

3. A maximum requires $A'(w)=0$, so the stated dimensions are not assumed optimal without that check.

=> FINAL: $A(w)=w(12-w)$; $A(9)=27$; check $A'(w)=0$ (check units/interpretation)